

# AKHILESH KUMAR MISHRA

## SOFTWARE DEVELOPMENT ENGINEER | MICROSERVICES · DISTRIBUTED SYSTEMS · FULL STACK

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### PROFILE

Software engineer, 3+ years building distributed systems for national e-governance and FinTech — 2 million+ users and 500,000+ transactions a day across Spring Boot, FastAPI and Node.js services. Cut p95 latency 45% across MySQL and PostgreSQL and 40% on MongoDB, lifted message throughput 35% on Kafka, and put every service behind Prometheus and Grafana. Onboarded 4 engineers and moved the team onto Docker-based CI/CD.

### CORE TECHNOLOGY

**Languages:** Java, Python, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

**Backend & Architecture:** Spring Boot, Django, FastAPI, Node.js, Express.js, React.js, Next.js, microservices, REST APIs, idempotency, rate limiting, WebSockets

**Data & Messaging:** MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, Celery, Hibernate/JPA, sharding, partitioning, query tuning

**Testing, DevOps & Observability:** pytest, JUnit, Jest, Docker, Kubernetes, GitHub Actions CI/CD, Linux, Jira, Prometheus, Grafana

### EXPERIENCE

#### Software Development Engineer | CSC e-Governance Services India Ltd. — India's national digital services platform, 2M+ users

Ministry of Electronics & Information Technology, Govt. of India  
12/2023 – PRESENT | New Delhi, India

- Sustained **500,000+ daily transactions** at a **99% success rate** across AEPS, MATM and UPI, by making the settlement path idempotent with request keys, backoff retries and automated reconciliation.
- Cut p95 read/write latency **45%** across **MySQL** and **PostgreSQL** and **40%** on **MongoDB**, by sharding and partitioning the hot tables and tuning the slowest queries against their plans.
- Contained a bad release to one service instead of the whole platform for **2 million+ users**, by splitting monolithic backends into **Spring Boot**, **FastAPI** and **Node.js** microservices behind a versioned gateway.
- Lifted message throughput **35%**, by moving inter-service work onto **Apache Kafka**, **Celery** and **Redis** with dead-letter handling for poison messages.
- Blocked **99.9%** of unauthorized access attempts, by adding **JWT** rotation, **RBAC**, device binding and OTP rate limiting across every payment entry point.
- Replaced a manual release process across all **3** payment platforms, by containerising every service with **Docker**, shipping through **GitHub Actions** and putting each one on **Prometheus** and **Grafana** dashboards.
- Improved team onboarding efficiency **40%**, by onboarding **4 engineers** and writing the runbooks and service documentation they start from.

#### Software Engineer | Contract engagement — field-workforce logistics, 1,000+ agents

Real-Time Telemetry & Geofencing Platform  
06/2023 – 11/2023 | Remote, India

- Engineered a workforce telemetry platform for **1,000+ field agents** end to end — **React.js** front end, **FastAPI** backend, AWS deployment.
- Held **99.9% uptime** while processing **10,000+ concurrent** GPS pings, through a **FastAPI**, **Celery** and **Redis** ingestion pipeline.
- Brought dashboard load time down **30%**, by building the live map in **React.js** with **Ant Design** over a WebSocket feed instead of polling.

### SELECTED PROJECTS

#### DigiPay Developer Platform

FastAPI / React / TypeScript / Docker / npm workspaces

Monorepo platform pairing a FastAPI service — versioned routers, correlation-ID tracing, auth and rate-limit middleware — with published SDK packages and embeddable React widgets. [github.com/GitHub-akhilesh/Django\\_apis\\_digipay](https://github.com/GitHub-akhilesh/Django_apis_digipay)

#### Polyglot Microservices Task Platform

Django / Flask / FastAPI / React / React Native / Kubernetes

Django, Flask and FastAPI services orchestrated under Docker Compose with Kubernetes manifests, fronted by a React web client and a React Native mobile app. [github.com/GitHub-akhilesh/To-do-list-app](https://github.com/GitHub-akhilesh/To-do-list-app)

#### Transaction Orchestration & Ledger Platform

Spring Boot / FastAPI / Kafka / PostgreSQL / Redis

Microservice platform orchestrating AEPS, MATM and UPI settlement with a transactional wallet ledger, idempotent retries, automated reconciliation and Kafka event fan-out.

#### LabX Design System

Python / System Design / Technical Documentation / GitHub Pages

Industrial data and visual automation project documented end to end — business case, HLD, LLD, data-flow and database design — and published as a live site. [github.com/GitHub-akhilesh/labx-design](https://github.com/GitHub-akhilesh/labx-design)

### EDUCATION

#### Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 - 2023